BRIGHT TUITION CLASSES

Time : 3 hrs

Date:14/12//25

Time: 11:30 AM to 2:30 PM

NEET

M.M. : 720

STD : 11

Important Instructions :

1. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology**
   50 questions in each subject are divided into **two sections (A and B)** as per details given below:
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos.-1 to 35, 51 to 85, 101 to 135 and 151 to 185). All Questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos. – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section-B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
2. Each question carries **4 marks**. For each correct response, the candidate will get **4 marks**. For each incorrect response, **1 mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue / Black Ball point Pen only** for writing particulars on this page / marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must handover the Answer Sheet (ORIGINAL and OFFICE Copy)** to the **Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
7. Each candidate must show on-demand his/her Admit Card to the Invigilator.
8. No candidate, without special permission of the Centre Superintendent or Invigilator, would leave his/her seat.
9. Use of Electronic/Manual Calculator is prohibited.
10. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
11. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
12. The candidates will write the Correct Test Booklet Code as given in the Test Booklet / Answer Sheet in the Attendance Sheet.

Name of the Candidate (In CAPITALS) : .....

Roll Number : .....

OMR Bar Code Number : .....

Candidate's Signature : ..... Invigilator's Signature .....

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Bright Tuition Classs

Subject : Physics
Standard : 11
Total Mark : 180

Std 11 th NEET 14 th December

Paper Set : 1
Date : 14-12-2025
Time : 3H:0M

Physics - Section A (MCQ)

(1) When a block of mass M is suspended by a long wire of length L, the length of the wire become $(L+l)$. The elastic potential energy stoped in the extended wire is :

(A) Mgl (B) MgL

(C) $\frac{1}{2}Mgl$ (D) $\frac{1}{2}$MgL

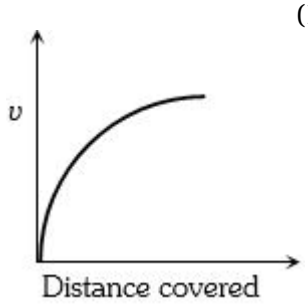
(2) A boy has $60\,kg$ weight. He wants to swim in a river with the help of a wooden log. If relative density of wood is 0.6, what is the minimum volume of wooden log? (density of river water is $1000\,kg/m^3$)

(A) $0.66\,m^3$ (B) $150\,m^3$

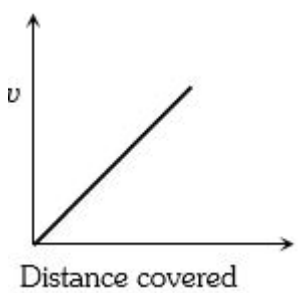
(C) $\frac{3}{1}m^3$ (D) $\frac{3}{20}m^3$

(3) A lead shot of $1mm$ diameter falls through a long column of glycerine. The variation of its velocity v. with distance covered is represented by

(A)



(B)



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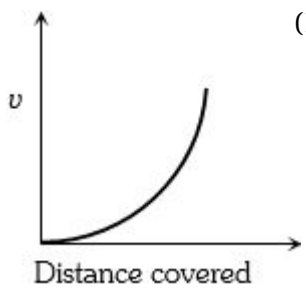
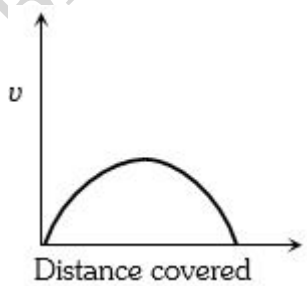


(D)



(4) The amount of energy required to form a soap bubble of radius $2\,cm$ from a soap solution is nearly ..... $\times 10^{-4}\,J$: (surface tension of soap solution $= 0.03\,N\,m^{-1}$)

(A) 50.1 (B) 30.16

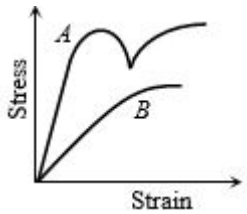
(C) 5.06 (D) 3.01

(5) If work done in increasing the size of a soap film from $10\,cm \times 6\,cm$ to $10\,cm \times 11\,cm$ is $2 \times 10^{-4}\,J$, then the surface tension is

(A) $2 \times 10^{-2}\,Nm^{-1}$ (B) $2 \times 10^{-4}\,Nm^{-1}$

(C) $2 \times 10^{-6}\,Nm^{-1}$ (D) $2 \times 10^{-8}\,Nm^{-1}$

(6) The diagram shows stress v/s strain curve for the materials A and B. From the curves we infer that



(A) A is brittle but B is ductile

(B) A is ductile and B is brittle

(C) Both A and B are ductile

(D) Both A and B are brittle

(7) Bernoulli's principle is based on the law of conservation of

(A) mass (B) energy

(C) angular momentum (D) linear momentum

(8) An elastic material of Young's modulus Y is subjected to a stress S. The elastic energy stored per unit volume of the material is

(A) $\frac{2Y}{S^2}$ (B) $\frac{S^2}{2Y}$

(C) $\frac{S}{2Y}$ (D) $\frac{S^2}{Y}$

(9) A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is

(A) $\frac{MgL}{A(L_1-L)}$ (B) $\frac{MgL_1}{AL}$

(C) $\frac{Mg(L_1-L)}{AL}$ (D) $\frac{MgL}{AL_1}$

(10) A stretched rubber has

(A) Increased kinetic energy (B) Increased potential energy

(C) Decreased kinetic energy (D) Decreased potential energy

(11) Excess pressure inside a soap bubble is three times that of the other bubble, then the ratio of their volumes will be

(A) $1:3$ (B) $1:9$

(C) $1:27$ (D) $1:81$

(12) $Assertion$: The pressure of water reduces when it flows from a narrow pipe to a wider pipe.
$Reason$: Since for wider pipe area is large, so flow of speed is small and pressure also reduces proportionately.

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(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(13) A container of height $10\,cm$ is filled with water. There is a hole at bottom. Find the pressure difference between points at top and bottom.

(A) $1000\,Pa$ (B) zero

(C) $1\,Pa$ (D) $100\,Pa$

(14) The spherical shape of rain-drop is due to

(A) Density of the liquid (B) Surface tension

(C) Atmospheric pressure (D) Gravity

(15) The normal density of a material is ρ and its bulk modulus of elasticity is K. The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be

(A) $\frac{\rho K}{P}$ (B) $\frac{\rho P}{K}$

(C) $\frac{K}{\rho P}$ (D) $\frac{PK}{\rho}$

(16) $Assertion$: Stress is the internal force per unitarea of a body.
$Reason$: Rubber is more elastic than steel.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If the Assertion is correct but Reason is incorrect.

(C) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(D) If both the Assertion and Reason are incorrect.

(17) The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :

(A) $\frac{Mg}{2}$ (B) Mg

(C) $\frac{3}{2}$Mg (D) 2Mg

(18) A body is just floating on the surface of a liquid. The density of the body is same as that of the liquid. The body is slightly pushed down. What will happen to the body

(A) It will slowly come back to its earlier position

(B) It will remain submerged, where it is left

(C) It will sink

(D) It will come out violently

(19) There are two wire of same material and same length while the diameter of second wire is two times the diameter of first wire, then the ratio of extension produced in the wires by applying same load will be

(A) $1:1$ (B) $2:1$

(C) $1:2$ (D) $4:1$

(20) The lift of an air plane is based on

(A) Torricelli's theorem

(B) Bernoulli's theorem

(C) Law of gravitation

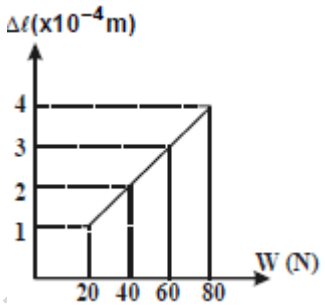
(D) Conservation of linear momentum

(21) A hydraulic automobile lift is designed to lift cars with a maximum mass of $3000\,kg$. The area of cross section a of piston carrying the load is $425\,cm^2$. What is the maximum pressure () would smaller piston have to bear ?

(A) $15.82 \times 10^5 Pa$ (B) $1.12 \times 10^5 Pa$

(C) $2.63 \times 10^5 Pa$ (D) $6.92 \times 10^5 Pa$

(22) The adjacent graph shows the extension (Δl) of a wire of length $1\,m$ suspended from the top of a roof at one end and with a load W connected to the other end. If the cross-sectional area of the wire is $10^{-6}\,m^2$, calculate the Young's modulus of the material of the wire.



(A) $2 \times 10^{11}\,N/m^2$ (B) $2 \times 10^{-11}\,N/m^2$

(C) $3 \times 10^{-12}\,N/m^2$ (D) $2 \times 10^{-13}\,N/m^2$

(23) Two small spherical metal balls, having equal masses, are made from materials of densities ρ_1 and $\rho_2\,(\rho_1 = 8\rho_2)$ and have radii of 1 mm and 2 mm, respectively. They are made to fall vertically (from rest) in a viscous medum whose coefficient of viscosity equals η and whose denstry is $0.1\rho_2$. The ratio of their terminal velocitites would be

(A) $\frac{79}{72}$ (B) $\frac{19}{36}$

(C) $\frac{39}{72}$ (D) $\frac{79}{36}$

(24) A rectangular film of liquid is extended from $(4\ cm \times 2\,cm)$ to $(5\ cm \times 4\,cm)$. If the work done is $3 \times 10^{-4}\,J$, the value of the surface tension of the liquid is ..... Nm^{-1}

(A) 0.2 (B) 8.0

(C) 0.250 (D) 0.125

(25) A soap bubble in vacuum has a radius of $3\,cm$ and another soap bubble in vacuum has a radius of $4\,cm$. If the two bubbles coalesce under isothermal condition, then the radius of the new bubble is ..... cm

(A) 2.3 (B) 4.5

(C) 5 (D) 7

(26) In a $U-$tube as shown in figure, water and oil are in the left side and right side of the tube respectively. The helghts from the bottom for water and oil columns are 15 cm and 20 cm respectively. The density of the oil is.....kg/m^3
[take $\rho_{water} = 1000$ kg/m^3]

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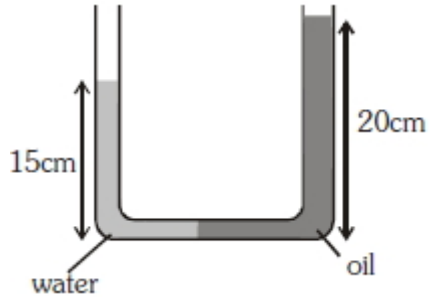


(A) 1200 (B) 750

(C) 1000 (D) 1333

(27) *Assertion* : In a pressure cooker the water is brought to boil. The cooker is then remove from the stove. Now on removing the lid of the pressure cooker, the water starts boiling against.
Reason : The impurities in water bring down its boiling point

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(28) *Assertion* : A bubble comes from the bottom of a lake to the top.
Reason : Its radius increases.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(29) A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking g $= 10$ m/s^2, the rate of flow of water through the open hole would be nearly ..... $\times 10^{-6}\ m^3/s$

(A) 12.6 (B) 8.9

(C) 2.23 (D) 6.4

(30) *Assertion* : Strain causes the stress in an elastic body.
Reason : An elastic rubber is more plastic in nature.

(A) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(31) *Assertion* : Machine parts are jammed in winter.
Reason : The viscosity of lubricant used in machine parts increase at low temperatures.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(32) The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are 8×10^8 N m^{-2} and 2×10^{11} N m^{-2}, is:

(A) 0.4 mm (B) 40 mm

(C) 8 mm (D) 4 mm

(33) Three liquids of equal masses are taken in three identical cubical vessels A, B and C. Their densities are ρ_A, ρ_B and ρ_C. The force exerted by the liquid on the base of the cubical vessel is

(A) Maximum in vessel C (B) Minimum in vessel C

(C) The same in all vessels (D) Maximum in vessel A

(34) The terminal velocity of a copper ball of radius $5\,mm$ falling through a tank of oil at room temperature is $10\,cm\,s^{-1}$. If the viscosity of oil at room temperature is $0.9\,kg\,m^{-1}s^{-1}$, the viscous drag force is :

(A) $8.48 \times 10^{-3}\,N$ (B) $8.48 \times 10^{-5}\,N$

(C) $4.23 \times 10^{-3}\,N$ (D) $4.23 \times 10^{-6}\,N$

(35) Water is flowing with a velocity of $2\,m/s$ in a horizontal pipe where cross-sectional area is $2 \times 10^{-2}\,m^2$ at pressure $4 \times 10^4\,pascal$. The pressure at cross-section of area $0.01\,m^2$ in pascal will be

(A) 32 (B) 3.4

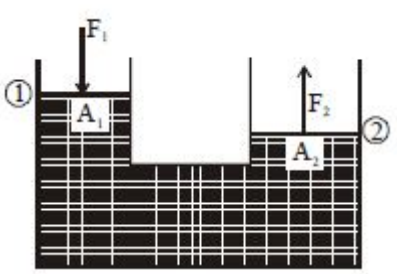
(C) 3.4×10^4 (D) 3.4×10^5

(36) The surface of water in a water tank of cross section area $750\,cm^2$ on the top of a house is $hm.$ above the tap level. The speed of water coming out through the tap of cross section area $500\,mm^2$ is $30\,cm/s$. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3} m/s$. The value of x will be .....

(A) 3 (B) 4

(C) 6 (D) 2

(37) The diagram below shows a hydraulic lift.A force is applied at *side* 1 and an output force is generated at *side* 2. Which of the following is true?



(A) The force at *side* 1 is greater than the force at *side* 2

(B) The force at *side* 1 is less than the force at *side* 2

(C) The pressure at *side* 1 is greater than the force at *side* 2

(D) The pressure at *side* 1 is less than the pressure at *side* 2

(38) A piston of cross-section area $100\,cm^2$ is used in a

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hydraulic press to exert a force of $107\,dynes$ on the water. The cross-sectional area of the other piston which supports an object having a mass $2000\,kg$. is

(A) $100\,cm^2$ (B) $109\,cm^2$

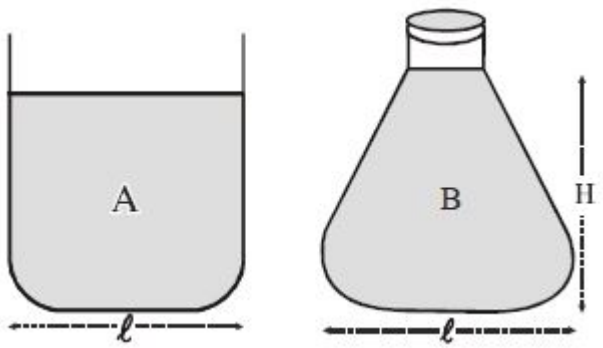
(C) $2 \times 10^4\,cm^2$ (D) $2 \times 10^{10}\,cm^2$

(39) The blades of a windmill sweep out a circle of area A. If the wind flows at a velocity b perpendicular to the circle, then the mass of the air of density ρ passing through it in time t is

(A) $Av\rho t$ (B) $2Av\rho t$

(C) $Av^2\rho t$ (D) $\frac{1}{2}Av\rho t$

(40) Two vessels A and B have the same base area and contain water to the same height, but the mass of water in A is four times that in B. The ratio of the liquid thrust at the base of A to that at the base of B is



(A) $4:1$ (B) $2:1$

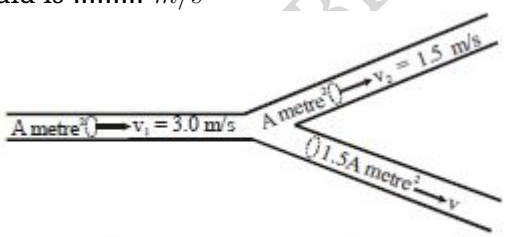
(C) $1:1$ (D) $16:1$

(41) Water is moving with a speed of $5.18ms^{-1}$ through a pipe with a cross-sectional area of $4.20cm^2$. The water gradually descends $9.66m$ as the pipe increase in area to $7.60cm^2$. The speed of flow at the lower level is ..... ms^{-1}

(A) 3.0 (B) 5.7

(C) 3.82 (D) 2.86

(42) An incompressible liquid flows through a horizontal tube as shown in the figure. Then, the velocity v of the fluid is ..... m/s



(A) 3 (B) 1.5

(C) 1 (D) 2.25

(43) Two water pipes of diameters $2cm$ and $4cm$ are connected with the main supply line. The velocity of flow of water in the pipe of $2cm$ diameter is

(A) 4 times that in the other pipe

(B) $\frac{1}{4}$ times that in the other pipe

(C) 2 times that in the other pipe

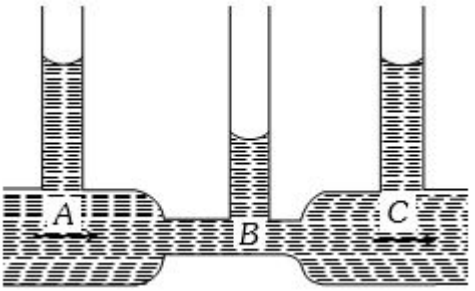
(D) $\frac{1}{2}$ times that in the other pipe

(44) A hosepipe directs a horizontal jet of water moving with velocity of $20\,m/s$ on a vertical wall. The cross-sectional area of jet is $10^{-3}\,m^2$. If density of water is $1000\,kg/m^3$, then force acting on wall is ..... N [Assuming water comes to rest after striking wall]

(A) 200 (B) 400

(C) 600 (D) 800

(45) In the following fig. is shown the flow of liquid through a horizontal pipe. Three tubes A, B and C are connected to the pipe. The radii of the tubes A, B and C at the junction are respectively $2cm, 1cm$ and $2cm$. It can be said that the



(A) Height of the liquid in the tube A is maximum

(B) Height of the liquid in the tubes A and B is the same

(C) Height of the liquid in all the three tubes is the same

(D) Height of the liquid in the tubes A and C is the same

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Bright Tuition Class

Subject : Chemistry
Standard : 11
Total Mark : 180

11th NEET Thermodynamics

Paper Set : 1
Date : 14-12-2025
Time : 0H:0M

Chemistry - Section A (MCQ)

(46) The standard heat of formation, in kcal/mol of Ba^{2+} is
[Given : standard heat of formation of SO_4^{2-} ion (aq) $= -216$ kcal/mol,
Standard heat of crystallisation of $BaSO_4(s) = -4.5$ kcal/mol, standard heat of formation of $BaSO_4(s) = -349$ kcal/ mol]
(A) -128.5 (B) -133.0
(C) $+133.0$ (D) $+220.5$

(47) For irreversible expansion of an ideal gas under isothermal condition, the correct option is :
(A) $\Delta U = 0, \Delta S_{total} = 0$ (B) $\Delta U \neq 0, \Delta S_{total} \neq 0$
(C) $\Delta U = 0, \Delta S_{total} \neq 0$ (D) $\Delta U \neq 0, \Delta S_{total} = 0$

(48) One mole of an ideal gas at $300\,K$ is expanded isothermally from $1\,L$ to $10\,L$ volume. ΔU for this process is ..... j (Use $R = 8.314\,J\,K^{-1}mol^{-1}$)
(A) 1260 (B) 2520
(C) 5040 (D) 0

(49) Which amongst the following options is the correct relation between change in enthalpy and change in internal energy?
(A) $\Delta H + \Delta U = \Delta nR$
(B) $\Delta H = \Delta U - \Delta n_9 RT$
(C) $\Delta H = \Delta U + \Delta n_g RT$
(D) $\Delta H - \Delta U = -\Delta nRT$

(50) Which of the following statements is correct for the spontaneous adsorption of a gas?
(A) ΔS is negative and, therefore ΔH should be highly positive.
(B) ΔS is negative and therefore, ΔH should be highly negative.
(C) ΔS is positive and therefore, ΔH should be negative.
(D) ΔS is positive and therefore, ΔH should also be highly positive.

(51) Under isothermal and reversible conditions, the term 'free energy' in thermodynamics signifies
(A) Non-expansion work done by the system
(B) Expansion work done by the system
(C) Non-expansion work done on the system
(D) Expansion work done on the system

(52) For a sample of perfect gas when its pressure is changed isothermally from p_i to p_f, the entropy change is given by
(A) $\Delta S = nR\,ln\left(\frac{p_f}{p_i}\right)$ (B) $\Delta S = nR\,ln\left(\frac{p_i}{p_f}\right)$
(C) $\Delta S = nRT\,ln\left(\frac{p_f}{p_i}\right)$ (D) $\Delta S = RT\,ln\left(\frac{p_i}{p_f}\right)$

(53) The correct thermodynamic conditions for the spontaneous reaction at all temperatures is
(A) $\Delta H > 0$ and $\Delta S < 0$ (B) $\Delta H < 0$ and $\Delta S < 0$
(C) $\Delta H < 0$ and $\Delta S = 0$ (D) $\Delta H < 0$ and $\Delta S > 0$

(54) An ideal gas expands isothermally from 10^{-3} m^3 to 10^{-2} m^3 at 300 K against a constant pressure of 10^5 Nm^{-2}. The work done on the gas is
(A) 270 kJ (B) -900 J
(C) $+900$ kJ (D) -900 kJ

(55) Match List-I with List-II.

List $-I$ (Process)	List $-II$ (Conditions)
A. Isothermal process	I. No heat exchange
B. Isochoric process	II. Carried out at constant temperature
C. Isobaric process	III. Carried out at constant volume
D. Adiabatic process	IV. Carried out at constant pressure

Choose the correct answer from the options given below:
(A) $A - IV, B - II, C - III, D - I$
(B) $A - I, B - II, C - III, D - IV$
(C) $A - II, B - III, C - IV, D - I$
(D) $A - IV, B - III, C - II, D - I$

(56) In which of the following processes entropy increases?
A. A liquid evaporates to vapour.
B. Temperature of a crystalline solid lowered from 130 K to 0 K.
C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$
D. $Cl_{2(g)} \rightarrow 2Cl_{(g)}$
Choose the correct answer from the options given below:
(A) A, B and D (B) A, C and D
(C) C and D (D) A and C

(57) In which case change in entropy is negative ?

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(A) Evaporation of water

(B) Expansion of a gas at constant temperature

(C) Sublimation of solid to gas

(D) $2H(g) \rightarrow H_2(g)$

(58) Of the following, the largest value of entropy at $25°C$ and $1\,atm$ is that of

(A) H_2 (B) C_2H_6

(C) C_2H_2 (D) CH_4

(59) If for a certain reaction $\Delta_r H$ is $30\ kJ\ mol^{-1}$ at $450\ K$, the value of $\Delta_r S$ (in $JK^{-1}mol^{-1}$) for which the same reaction will be spontaneous at the same temperature is

(A) -70 (B) 70

(C) -33 (D) 33

(60) At standard conditions, if the change in the enthalpy for the following reaction is $-109\ kJmol^{-1}$

$H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$

Given that bond energy of H_2 and Br_2 is $435\ kJmol^{-1}$ and $192\ kJmol^{-1}$, respectively, what is the bond energy (in $kJmol^{-1}$) of HBr?

(A) 259 (B) 368

(C) 736 (D) 518

(61) Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : In equation $\Delta_r G = -nFE_{\text{cell}}$ value of $\Delta_r G$ depends on n.

Reasons R : E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.

In the light of the above statements, choose the correct answer from the options given below

(A) A is false but R is true

(B) Both A and R are true and R is the correct explanation of A

(C) Both A and R are true and R is NOT the correct explanation of A

(D) A is true but R is false

(62) A reaction having equal energies of activation for forward and reverse reactions has

(A) $\Delta H = 0$ (B) $\Delta H = \Delta G = \Delta S = 0$

(C) $\Delta S = 0$ (D) $\Delta G = 0$

(63) A gas is allowed to expand in a well insulated container against a constant external pressure of $2.5\ atm$ from an initial volume of $2.50\ L$ to a final volume of $4.50\ L$. The change in internal energy ΔU of the gas in joules will be ..... J

(A) -500 (B) -505

(C) $+505$ (D) 1136.25

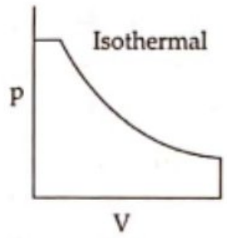
(64) For a given reaction, $\Delta H = 35.5\ kJ\ mol^{-1}$ and $\Delta S = 83.6\ J\ K^{-1}\ mol^{-1}$. The reaction is spontaneous at (Assume that ΔH and ΔS do not vary with temperature.)

(A) $T > 425\ K$ (B) all temperatures

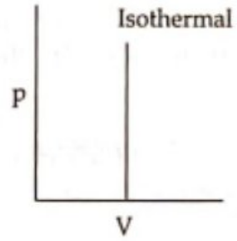
(C) $T > 298\ K$ (D) $T < 425\ K$

(65) Which of the following $p - V$ curve represents maximum work done?

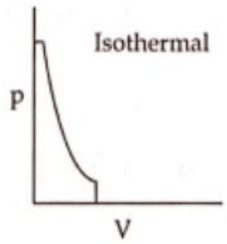
(A)



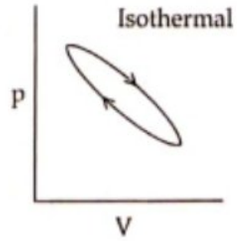
(B)



(C)



(D)



(66) The heat of combustion of carbon to CO_2 is $-393.5\ kJ/mol$. The heat released upon formation of $35.2\ g$ of CO_2 from carbon and oxygen gas is

(A) $+315\,kJ$ (B) $-630\,kJ$

(C) $-3.15\,kJ$ (D) None of the above

(67) The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20atmosphereto10atmosphere is

(Given $R = 2.0\text{calK}^{-1}\ \text{mol}^{-1}$)

(A) -413.14 calories (B) 413.14 calories

(C) 100 calories (D) 0 calorie

(68) Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas?

(A) $C_P + C_V = R$ (B) $C_P - C_V = R$

(C) $C_P = RC_V$ (D) $C_V = RC_P$

(69) At $25°C$ and $1\,atm$ pressure, the enthalpy of combustion of benzene$_{(l)}$ and acetylene$_{(g)}$ are $-3268\,kJ\,mol^{-1}$ and $-1300\,kJ\,mol^{-1}$, respectively. The change in enthalpy for the reaction

$3C_2H_{2(g)} \rightarrow C_6H_{6(l)}$, is ..... $kJ\,mol^{-1}$

(A) $+324$ (B) $+632$

(C) -632 (D) -324

(70) For the reaction

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)$ the magnitude of the standard molar free energy change, $\Delta_r G_m^\circ = -$ ..... kJ (Round off to the Nearest Integer).

$$\left[\begin{array}{ll} E^o_{Fe^{2+}/Fe(s)} = -0.440V; & E^o_{Fe^{3+}/Fe(s)} = -0.036V \\ E^o_{I_2/2I^-} = 0.539V; & F = 96500\,C \end{array}\right]$$

(A) 49 (B) 36

(C) 55 (D) 46

(71) Data given for the following reaction is as follows:

$FeO_{(0)} + C_{(gaplike)} \longrightarrow Fe_{(0)} + CO_{(g)}$

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Substance	ΔH° ($kJmol^{-1}$)	ΔS° ($Jmol^{-1}\ K^{-1}$)
$FeO_{(s)}$	-266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe_{(s)}$	0	27.28
$CO_{(g)}$	-110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is ..... .
(Integer answer)

(A) 964 (B) 864

(C) 96.4 (D) 9.64

(72) The reaction of cyanamide, $NH_2CN_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be $-742.24\,kJ\,mol^{-1}$. The magnitude of ΔH_{298} for the reaction
$NH_2CN_{(s)} + \frac{3}{2}O_2(g) \rightarrow N_{2(g)} + O_2(g) + H_2O_{(l)}$
is ..... kJ. (Rounded off to the nearest integer)
[Assume ideal gases and $R = 8.314\,J\,mol^{-1}K^{-1}$]

(A) 343 (B) 741

(C) 783 (D) 561

(73) An average person needs about 10000 kJ energy per day. The amount of glucose (molar mass $= 180.0$ g mol^{-1}) needed to meet this energy requirement is ..... g.
(Nearest integer)
(Use : $\Delta_c H$(glucose) $= -2700$ kJ mol^{-1})

(A) 700 (B) 667

(C) 845 (D) 321

(74) For the reaction taking place in the cell:
$Pt(s)\,|H_2(g)|\,H^+(aq)\|Ag^+(aq)\,|\,Ag(s)$
$E^0_{Cell} = +0.5332V$.
The value of $\Delta_f G^0$ is $kJ\,mol^{-1}$. (in nearest integer)

(A) 5 (B) 96

(C) 51 (D) 57

(75) For independent process at 300 K.

Process	$\Delta H/kJ\,mol^{-1}$	$\Delta S/JK^{-1}$
A	-25	-80
B	-22	40
C	25	-50
D	22	20

The number of non-spontaneous process from the following is .....

(A) 4 (B) 6

(C) 2 (D) 8

(76) Butane reacts with oxygen to produce carbon dioxide and water following the equation given below
C_4H_{10}(g) + $\frac{13}{2}O_2$(g) $\rightarrow 4CO_2$(g) + $5H_2O$(l) If 174.0 kg of butane is mixed with 320.0 kg of O_2, the volume of water formed in litres is .....(Nearest integer)[Given : (a) Molar mass of C, H, O are 12, 1, 16 g mol^{-1} respectively, (b) Density of water $= 1$ g mL^{-1}]

(A) 123 (B) 248

(C) 138 (D) 158

(77) For a dimerization reaction,
$2A(g) \rightarrow A_2(g)$
at
$298\,K, \Delta U^\ominus, = -20\,kJ\,mol^{-1}, \Delta S\odot = -30\,JK^{-1}\,mol^{-1}$,
then the $\Delta G^\ominus$ will be.....J

(A) 13536.6 (B) -13537.6

(C) -13535.5 (D) 13530.2

(78) One mole of an ideal gas at $350\,K$ is in a $2.0\,L$ vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from $2.0\,L$ to $3.0\,L$ against a constant pressure of $4\,atm$. The change in entropy of the surroundings (ΔS) is ..... $J\,K^{-1}$ (Nearest integer)
Given : $R = 8.314\,J\,K^{-1}\,Mol^{-1}$.

(A) 4 (B) 3

(C) 5 (D) 6

(79) The following statements relate to the adsorption of gases on a solid surfuce. Identify the incorrect statement among them

(A) Enthalpy of adsorption is negative

(B) Energy appears as heat

(C) On adsorption, the residual forces on the surface are increased

(D) Entropy of adsorption is negative

(80) A gas (Molar mass $= 280\,g\,mol^{-1}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from $298.0K$ to $298.45\,K$. If the heat capacity of calorimeter is $2.5\,kJ\,K^{-1}$ and enthalpy of combustion of gas is $9\,kJ\,mol^{-1}$ then amount of gas burnt is .....g.
(Nearest Integer)

(A) 35 (B) 34

(C) 33 (D) 32

(81) The standard entropy change for the reaction
$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is $-550\,JK^{-1}$ at $298\,K$.
[Given : The standard enthalpy change for the reaction is $-165\,kJ\,mol^{-1}$. The temperature in K at which the reaction attains equilibrium is ... .
(Nearest Integer)

(A) 30 (B) 300

(C) 210 (D) 670

(82) While performing a thermodynamics experiment, a student made the following observations,
$HCl + NaOH \rightarrow NaCl + H_2O\Delta H = -57.3\,kJ\,mol^{-1}$
$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
$(\Delta H = -55.3\,kJ\,mol^{-1})$ The enthalpy of ionization of CH_3COOH as calculated by the student is $kJmol^{-1}$.
(nearest integer)

(A) 5 (B) 2

(C) 4 (D) 3

(83) ΔU is equal to

(A) Isochoric work (B) Isobaric work

(C) Adiabatic work (D) Isothermal work

(84) For an ideal solution of two components A and B, which of the following is true?

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(A) $\Delta H_{mixing} < 0$ (zero)

(B) $\Delta H_{mixing} > 0$ (zero)

(C) $A - B$ interaction is stronger than $A - A$ and $B - B$ interactions

(D) $A - A, B - B$ and $A - B$ interactions are identical

(85) Among the following the number of state variable is .....
Internal energy (U); Volume (V); Heat (q); Enthalpy (H)

(A) 2 (B) 3

(C) 4 (D) 6

(86) The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are $55.0\ kJ\ mol^{-1}$ and $175.0 J\ K\ mol$ respectively. The standard free energy change for this reaction at $25°C$ in $J\ mol^{-1}$ is .....(Nearest integer)

(A) 2850 (B) 2950

(C) 3065 (D) 4025

(87) For complete combustion of methanol
$CH_3OH(1) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(1)$
the amount of heat produced as measured by bomb calorimeter is $726\,kJ\,mol^{-1}$ at $27°C$. The enthalpy of combustion for the reaction is $-x\,kJ\,mol^{-1}$, where x is ..... (Nearest integer)
(Given : $R = 8.3\,J\,K^{-1}\,mol^{-1}$)

(A) 314 (B) 632

(C) 552 (D) 727

(88) When a $60\,W$ electric heater is immersed in a gas for $100\,s$ in a constant volume container with adiabatic walls, the temperature of the gas rises by $5°\,C$. The heat capacity of the given gas is ..... JK^{-1} (Nearest integer)

(A) 1100 (B) 1200

(C) 1000 (D) 1600

(89) The effect of temperature on spontaneity of reactions are represented as:

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	−	any T	Non spontaneous
(B)	+	+	low T	spontaneous
(C)	−	−	low T	Non spontaneous
(D)	−	+	any T	spontaneous

(A) (B) and (D) only (B) (A) and (D) only

(C) (B) and (C) only (D) (A) and (C) only

(90) $4.0\,L$ of an ideal gas is allowed to expand isothermally into vacuum until the total volume is $20\,L$. The amount of heat absorbed in this expansion is ..... $L\,atm$

(A) 10 (B) 20

(C) 0 (D) 1

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Subject : Bio
Standard : 11
Total Mark : 360

Class 11 NEET Practice Paper - 9

Paper Set : 1
Date : 14-12-2025
Time : 1H:0M

Biology - Section A (MCQ)

(91) Pyruvate dehydrogenase activity during aerobic respiration requires

(A) Magnesium (B) Calcium

(C) Iron (D) Cobalt

(92) Which hormone promotes internode/petiole elongation in deep water rice?

(A) $2,4-D$ (B) GA_3

(C) Kinetin (D) Ethylene

(93) Match the columns.

$Column-I$	$Column-II$
$(a)4Ccompound$	$(1)AcetylCoA$
$(b)2Ccompound$	$(2)Pyruvate$
$(c)5Ccompound$	$(3)Citricacid$
$(d)3Ccompound$	$(4)\alpha-Ketoglutaticacid$
	$(5)Malicacid$

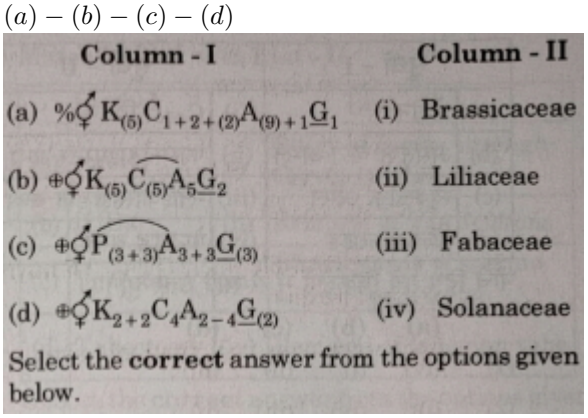
(A) $a-2, b-5, c-3, d-1$

(B) $a-5, b-1, c-4, d-2$

(C) $a-3, b-1, c-4, d-2$

(D) $a-5, b-3, c-1, d-2$

(94) Match Column $-I$ with Column $-II$.
$[Image]$
Select the correct answer from the options given below.
$(a)-(b)-(c)-(d)$



Column - I	Column - II
(a) $\%\,⚥\,K_{(5)}C_{1+2+(2)}A_{(9)+1}\underline{G}_1$	(i) Brassicaceae
(b) $\oplus\,⚥\,K_{(5)}\overset{\frown}{C_{(5)}A}_5\underline{G}_2$	(ii) Liliaceae
(c) $\oplus\,⚥\,\overset{\frown}{P_{(3+3)}A}_{3+3}\underline{G}_{(3)}$	(iii) Fabaceae
(d) $\oplus\,⚥\,K_{2+2}C_4A_{2-4}\underline{G}_{(2)}$	(iv) Solanaceae

Select the **correct** answer from the options given below.

(A) $(iii)-(iv)-(ii)-(i)$ (B) $(i)-(ii)-(iii)-(iv)$

(C) $(i)-(ii)-(iii)-(iv)$ (D) $(iv)-(ii)-(i)-(iii)$

(95) In a flower, if a whorl member overlaps without any direction, than aestivation is known as :

(A) Vexillary (B) Imbricate

(C) Twisted (D) Valvate

(96) Which of the following was isolated and identified by Skoog and Miller?

(A) Coconut milk (B) Kinetin

(C) Zeatin (D) Yeast extract

(97) Leaves of dicotyledonous plants possess _____ venation, while _____ venation is the characteristic of most monocotyledons

(A) reticulate and parallel

(B) parallel and reticulate

(C) reticulate and perpendicular

(D) obliquely and parallel

(98) Choose incorrect pair.

(A) Gibberellic acid - leaf fall

(B) Cytokinin - delay senescence

(C) $IAA-$ help of prevent leaf drop at early stage

(D) $ABA-$ inhibit seed germination

(99) ..... is family of chilli.

(A) Fabaceae. (B) Liliaceae

(C) Solanaceae (D) Poaceae

(100) Respiration is an .....

(A) Endergonic process (B) Exergonic process

(C) A and B both (D) None of the above

(101) Seed develop from

(A) Ovules (B) Ovaries

(C) Anthers (D) Pistils

(102) A drupe develops in

(A) mango (B) wheat

(C) pea (D) tomato.

(103) Given below are two statements :
Statement I : In a floral formula $\oplus$ stands for zygomorphic nature of the flower, and $\underline{G}$ stands for inferior ovary.
Statement II : In a floral formula $\oplus$ stands for actinomorphic nature of the flower and $\underline{G}$ stands for superior ovary.
In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

(104) Choose incorrect pair
(Aestivation - example).

(A) Valvet - calotropis (B) Twisted - cotton

(C) Imbricate - Gulmohur (D) Vaxillary - china rose

(105) In anaerobic respiration in plants

(A) Oxygen is taken in

(B) Oxygen is given out

(C) Carbon dioxide is given out

(D) Carbon dioxide is taken in

(106) In ..... flowers are arranged in basipetal

(A) Cymose inflorescence

(B) Racemose inflorescence

(C) A and B correct

(D) A and B incorrect

(107) Read following statements regarding PGRs
A. Ethylene also promotes root growth and root hair formation,
B. Natural cytokinins are synthesised in regions where rapid cell division occurs
C.$2,4-D$, widely used to kill monocotyledonous weeds, does not affect mature dicotyledonous plants.

(A) only A is correct

(B) only B is incorrect

(C) only C is incorrect

(D) Both A and C are incorrect

(108) During ETS ubiquinone receives reducing equivalents via

(A) Complex IV (B) Complex III

(C) Complex V (D) Complex II

(109) Swollen lower end of leaf stalk is

(A) Petiole (B) Pulvinus

(C) Thalamus (D) Disc

(110) Which one of the following hormone play important role in seed development, maturation and dormancy ?

(A) ABA (B) IAA

(C) GA (D) CK

(111) A phyllotaxy with two or more leaves present on the same node is

(A) Opposite (B) Verticillate

(C) Whorled (D) Cyclic

(112) I. The increased growth per unit time is termedas ....A...
II. Abscisic acid .....B..... the stomata.
Choose the correct option and fill in the blanks.

(A) $A-$ Plasticity, $B-$ Close

(B) $A-$ Growth rate, $B-$ open

(C) $A-$ Plasticity, $B-$ open

(D) $A-$ Growth rate , $B-$ close

(113) Plasticity can be observed in the leaf of

(A) Lark spur (B) Neem

(C) Carrot (D) None

(114) Main function of leaf is

(A) Manufacture of food

(B) Nerve impulse conduction

(C) Increasing grandeur

(D) Exchange of gases

(115) The first phase in the breakdown of glucose in animal cell is

(A) Glycolysis

(B) Electron transport system

(C) Fermentation

(D) Krebs cycle

(116) Identify the correct group for perigynous flower.

(A) Guava, Mustard, china rose

(B) Rose, plum, peach

(C) Cucumber, brinjal, mustard

(D) Plum, brinjal, guava

(117) If the margins of sepals or petals overlap one another but not in any particular direction is called.

(A) Valvet (B) Twisted

(C) Imbricate (D) Vaxillary

(118) Identify false for dicot seed.

(A) Possess two cotyledons

(B) Food storage in cotyledons

(C) Absence of embryonic axis

(D) Possess seed coat

(119) Carbon dioxide liberates during

(A) Ascent of sap (B) Transpiration

(C) Photosynthesis (D) Respiration

(120) When proteins are respiratory substrutes the RQ would be about

(A) 0.7 (B) 1

(C) 4 (D) 0.9

(121) In a cymose inflorescence the main axis

(A) has unlimited growth

(B) bears a solitary flower

(C) has unlimited growth but lateral branches end in flowers

(D) terminates in a flower.

(122) Decarboxylation is not involved in

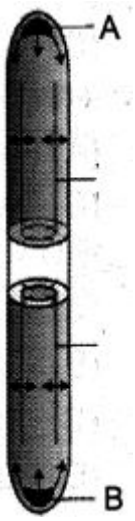
(A) Electron transport system

(B) Glycolysis

(C) Kreb's cycle

(D) Alcoholic fermentation

(123) Identify A and B in the given figure and choose the correct option.



(A) $A-$ Root apical meristem; $B-$ Shoot apical meristem

(B) $A-$ Shoot apical meristem; $B-$ Root apical meristem

(C) $A-$ Seed coat; $B-$ Radicle tissue

(D) $A-$ Radical tissue; $B-$ Root

(124) Find out the incorrect : –

(A) Valvate aestivation – Calotropis

(B) Twisted aestivation – Cotton

(C) Imbricate aestivation – Cassia

(D) Vaxillary aestivation – China rose

(125) Oxidation of succinate to fumerate in the Kreb's cycle is due to

(A) Loss of electron from it

(B) Removal of hydrogen from it

(C) Addition of oxygen to it

(D) None of the above

(126) $\underline{G}$ and $\overline{G}$ indicates respectively : –

(A) Epigynous and hypogynous flowers

(B) Superior ovary and inferior ovary

(C) Presence and absence of gynoecium

(D) Fused and free gynoecium

(127) Aestivation of corolla in Pea is

(A) Contorted (B) Valvate

(C) Imbricate (D) Vexillary

(128) Identify growth inhibitor hormone.

(A) Auxin (B) Cytokinine

(C) Geberellins (D) Ethylene

(129) It helps the overcome apical dominance ?

(A) IAA (B) Cytokinins

(C) Ethylene (D) NAA

(130) During anaerobic respiration in yeast

(A) Water and CO_2 are end products

(B) CO_2, C_2H_5OH and energy are end products

(C) H_2S, $C_6H_{12}O_6$ and energy are the end products

(D) H_2O, CO_2 and energy are the only end products

(131) Which effects are of gibberellins ?

(a) It help overcome the apical dominance

(b) Delay senescence, Thus the fruits can be left on the tree longer so as

(c) Increase in length of axis

(d) It is used to speed up the maltingprocess in brewing industries

(e) It induces parthenocarpy

(f) Promotes female flowers in cucumbers

(A) a, e, f (B) b, e, f

(C) d, b, c (D) a, c, d

(132) In the electron transport system in the inner mitochondria membrane, complex–I and IV are respectively : –

(A) NADH dihydrogenase and $FADH_2$

(B) $FADH_2$ and NADH dihydrogenase

(C) NADH dehydrogenase and cytochrome–c oxidase complex

(D) NADH dehydrogenase and ATP synthase

(133) In aerobic respiration

(A) Glucose is the only substrate

(B) Glucose is a preferred substrate

(C) Glucose cannot be used directly

(D) Any material which burns in air can be used as substrate

(134) Edible part of mango is

(A) Epicarp (B) Mesocarp

(C) Endocarp (D) Receptacle

(135) Read the following statements-

Statement–1 : A sterile stamen is called staminode.

Statement–2 : When stamens are attached to the petals, they are epipetalous as in brinjal

(A) Both statements are correct

(B) only statement 1 is correct

(C) only statement 2 is correct

(D) Both statements are incorrect

Biology - Section B (MCQ)

(136) ATP formation occurs through which of the following-

(A) Photophosphorylation

(B) Oxidative phosphorylation

(C) Substrate level phosphorylation

(D) All of these

(137) Which of the following is not an intermediate of Kreb's cycle

(A) Acetyl $Co-A$ (B) Citric acid

(C) Succinic acid (D) Lactic acid

(138) Which group of the following scientists discovered the EMP pathway of glycolysis

(A) Embden, Mayerhoff and Parnas

(B) Emerson, Hoffman and Peterson

(C) Embden, Morrison and Pitcher

(D) Avery, McLeod and McCarthy

(139) Which one of the following pairs, is not correctly matched?

(A) Gibberellic acid Leaf fall

(B) Cytokinin Cell division

(C) IAA Cell wall elongation

(D) Abscissic acid Stomatal closure

(140) If protein enters into amphibolic pathway, amino acid is converted into.....

(A) Pyrvic acid (B) Acetic acid

(C) Acetyl co $-A$. (D) $DHAP$

(141) The complex II of mitochondrial electron transport chain is also known as

(A) Cytochrome bc_1 (B) Succinate dehydrogenase

(C) Cytochrome c oxidase (D) NADH dehydrogenase

(142) Common immediate source of energy in cellular activity or Energy currency of the cell is

(A) DNA (B) ATP

(C) RNA (D) NAD

(143) Which of the following is a more accurate statement about respiration

(A) O_2 must always be available for respiration

(B) O_2 combines with carbon to form CO_2

(C) O_2 combines with hydrogen to form H_2O

(D) Air is inhaled and exhaled only from stomata

(144) Maize seed possess.

(A) Scutellum

(B) Aleurone layer

(C) Seed coat and fruit wall

(D) Given all

(145) Primary root and its branches constitute

(A) Adventitious root system

(B) Tap root system

(C) Fibrous roots

(D) Seminal roots

(146) First product of glycolysis is

(A) Glucose

(B) $BPGA$

(C) Glucose $-6-$ Phosphate

(D) Pyruvic acid

(147) When petiole bears leaflets at its tip, it is a

(A) Simple leaf

(B) Pinnate compound leaf

(C) Palmate compound leaf

(D) Isobilateral leaf

(148) Which type of sepal aestivation is found in Solanaceae family?

(A) Valvet (B) Vaxillary

(C) Imbricate (D) Twisted

(149) Which of the following is an example of actinomorphic flower?

(A) Cassia (B) Pisum

(C) Sesbania (D) Datura

(150) Fermentation is

(A) Anaerobic respiration

(B) Incomplete oxidation of carbohydrates

(C) Complete oxidation of carbohydrates

(D) None of the above

(151) Plants follow .....A.... pathways in response to environment or phases of life to form different kind of structures. This ability is called ....B..... Complete the given statement with the correct combination of options.

(A) $A-$ same; $B-$ elasticity

(B) $A-$ elasticity; $B-$ same

(C) $A-$ different; $B-$ plasticity

(D) $A-$ same; $B-$ plasticity

(152) Isomerization step takes place in

(A) Glycolysis

(B) Kreb's cycle

(C) Oxidative phosphorylation

(D) None of the above

(153) In which of the following step of glycolysis, ATP is utilized ?

(A) $DHAP \rightarrow$ biphosphoglyceric acid

(B) 3 phosphoglyceric acid $\rightarrow 2$ phosphoglycerate

(C) Fructose $-6-$ phosphate $\rightarrow$ fructose $1,6$ biphosphate

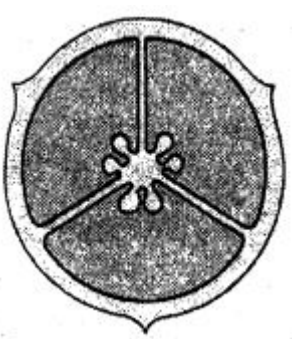
(D) Fructose $1,6-$ biphosphate $\rightarrow DHAP$ and $PGAL$

(154) Fungal pathogens which is responsible for foolish seeding disease of rice seeding?

(A) Aspergillus niger (B) Polysporum

(C) Trichophyton (D) Gibberella fujikuroi

(155) Placentation in the given figure is found in which plant?



(A) China rose (B) Dianthus

(C) Marigold (D) Argemone

(156) Match the correct pair.

Column $-I$	Column $-II$
(1) Imbricate	(P) Gulmohur
(2) Velvet	(Q) Cotton
(3) Twisted	(R) Caltrops
	(S) Pea

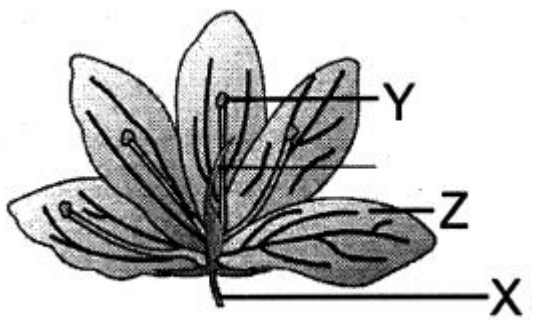
(A) $(1-P),(2-Q),(3-R)$

(B) $(1-Q),(2-S),(3-P)$

(C) $(1-P),(2-R),(3-Q)$

(D) $(1-S),(2-R),(3-P)$

(157) Identify X, Y, Z from the figure.
$X-Y-Z$



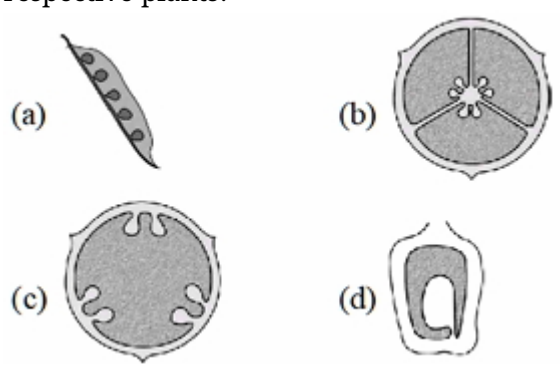
(A) Pedicel - gynoecium - calyx

(B) Pedicel - androecium - corolla

(C) Calyx - androecium - gynoecium

(D) Pedicel - corolla - androecium

(158) Cross sections of ovaries of four different plants are given below. Assign the placentations to their respective plants.



(A) $a=$ Pea, $b=$ Mustard, $c=$ Sunflower, $d=$ Chinarose

(B) $a=$ Pea, $b=$ Chinarose, $c=$ Mustard, $d=$ Sunflower

(C) $a=$ Chinarose, $b=$ Mustard, $c=$ Sunflower, $d=$ Pea

(D) $a=$ Mustard, $b=$ Sunflower, $c=$ Pea, $d=$ Chinarose

(159) Assertion: The cymose type of inflorescence has limited growth.
Reason: In cymose inflorescence the main axis terminates in a flower.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Both Assertion and Reason are false.

(160) Apocarpous ovary is found in : –

(A) Mustard and Tomato (B) Capsella and Radish

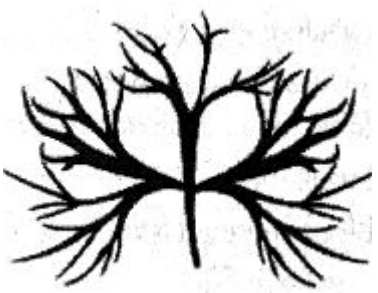
(C) Lotus and Rose (D) Potato and Tomato

(161) Formation of meristem intrafascicular cambiun and cork cambium from fully differentiated parenchyma cell. This is example of.

(A) Differentiation (B) Dediffentiation

(C) Redifferentiation (D) Senescenes.

(162) Identify figure.



(A) Larkspur (B) Neem

(C) butter cup (D) corriander

(163) Which of them is not an extrinsic factor?

(A) Light, O_2

(B) Temperature, CO_2

(C) Nutrient, water

(D) Growth regulator and genetic factor

(164) Kreb's cycle involves the formation of

(A) Lactic acid from glucose

(B) Change of pyruvic acid to energy transformation

(C) Pyruvic acid from glucose

(D) ATP from ADP

(165) The rosette habit of cabbage can be changed by inducing stem elongation by the application of.

(A) Ethylene (B) GA

(C) CK (D) ABA

(166) Which is not true for glycolysis

(A) End product is CO_2, H_2O

(B) Substiate level phosphorylation

(C) Production of ATP

(D) Expenditure of ATP

(167) In arithmetic growth, mitotic cell division, only one daughter cell continues to divide while the other .....

(A) Become dead

(B) Differentiates and matures

(C) Also continues divide

(D) Divides and suddenly dead

(168) $A-$ In plant or animal cell first respiratory phase is EMP.
$R-$ Other name of EMP is TCA.

(A) A and R both are correct.

(B) A and R both are incorrect.

(C) A is correct, R is incorrect.

(D) A is incorrect, R is correct.

(169) When 1 molecule of pyruvate is completely oxidized, how many moles of carbon dioxide are produced?

(A) 1 (B) 2

(C) 3 (D) 4

(170) The ..... is a scar on the seed coat through which the developing seeds were attached to the fruit.

(A) Hilum (B) Micropyle

(C) Plumule (D) Aleurone layer

(171) Respiratory quotient $(R.Q.)$ is

(A) $\frac{CO_2}{O_2}$ (B) $\frac{O_2}{CO_2}$

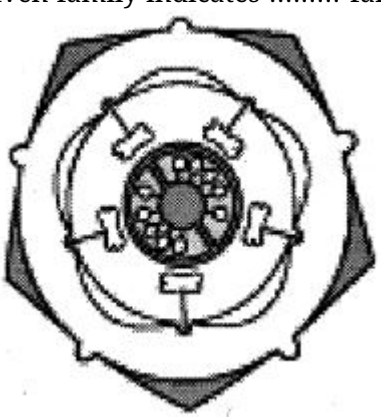
(C) $\frac{N_2}{CO_2}$ (D) $\frac{CO_2}{N_2}$

(172) Single shield shaped cotyledon of maize seed is called: –

(A) Coleorrhiza (B) Coleoptile

(C) Scutellum (D) Epiblast

(173) Given family indicates ..... family.



(A) Fabaceae (B) Solanaceae

(C) Liliaceae (D) Poaceae

(174) The branch system of the shoot bearing a group of flowers is called as

(A) Placentation (B) Venation

(C) Inflorescence (D) Phyllotaxy

(175) Horizontal growth of seedling, swelling of axis and apical hook formation in dicot seedling are associated with: –

(A) Photoperiodism (B) Vernalisation

(C) Ethylene (D) Cytokinin

(176) Oxidative phosphorylation occurs in

(A) Outer membrane of mitochondria

(B) Inner membrane of mitochondria

(C) Stroma of chloroplast

(D) Grana of chloroplast

(177) Which of the following is reverse of photosynthesis

(A) Respiration (B) Protein synthesis

(C) Fat synthesis (D) All the above

(178) If mitochondria is absent in mature RBC what will be the source of energy:

(A) TCA (B) ETS

(C) link reaction (D) Glycolysis

(179) Pyruvic acid is converted into a compound before formation of oxaloacetic acid in the citric acid cycle, this compound is

(A) Acetyl CoA (B) Acetoacetic acid

(C) Lactic acid (D) cisaconitic acid

(180) Tthe floral formula of

$$\oplus \; ⚥ \; K_{(5)} \; \overset{\frown}{C_{(5)} \; A_5} \; \underline{G}_{(2)}$$

(A) Petunia (B) Brassica

(C) Allium (D) Sesbania.